

1.2 Permutations and Combinations

Permutations

Permutation is an arrangement of objects wherein order is taken into account.

1. Permutation of objects taken all at a time.

$${}_nP_n = n! = n(n-1)(n-2) \dots (n-n+1)$$

Example 1.

In the word THEORY, how many different arrangements of letters can be made?

Solution:

$${}_6P_6 = 6!$$

$$= 6(5)(4)(3)(2)(1)$$

$$= 720 \text{ arrangements.}$$

2. Permutation of objects taken r at a time.

$${}_nP_r = n! / (n-r)!$$

Example 1.

In how many ways can a teacher assign the 4 key positions to organize a classroom activity to 7 equally qualified students?

Solution:

$$n = 7$$

$$r = 4$$

$${}_7P_4 = 7! / (7-4)!$$

$$= 7! / 3!$$

$$= (7)(6)(5)(4)(3!) / 3!$$

$$= 840 \text{ ways.}$$

3. Circular Permutation

A permutation in which one position must be fixed.

$$({n-1})P_{(n-1)} = (n-1)!$$

Example 1.

How many ways can 7 players be standing inside a circle with seven markings?

Solution:

$$({7-1})P_{(7-1)} = (7-1)!$$

$$= 6!$$

$$= (6)(5)(4)(3)(2)(1)$$

$$= 720 \text{ ways.}$$

4. Permutation of n objects not all distinct.

$$P = N!/(n_1! \cdot n_2! \cdot \dots \cdot n_k!)$$

where:

$$N = n_1 + n_2 + \dots + n_k$$

Example 1.

How many number of permutations of the letters in the word PROBABILITY can be made?

Solution:

$$P = 11!/(1! \cdot 1! \cdot 1! \cdot 2! \cdot 1! \cdot 2! \cdot 1! \cdot 1! \cdot 1!)$$

$$= (11)(10)(9)(8)(7)(6)(5)(2)(3)$$

$$= 9,979,200 \text{ permutations.}$$

Combinations

Combination is a selection of objects with no attention given to the order of the objects.

1. Combination of n objects taken all at the same time.

$${}_nC_n = 1$$

Example 1.

How many ways can 8 members form a committee of eight?

Solution:

$${}_8C_8 = 1 \text{ committee.}$$

2. Combination of n objects taken r at a time.

$${}_nC_r = n!/(n-r)!r!$$

Example 1.

The TIMES Organization is forming a group of seven to be made up of four from the males and three from the females. How many ways are there of selecting the group if seven nominees come from the males and six nominees come from the females?

Solution:

Males

Females

$$n = 7$$

$$n = 6$$

$$r = 4$$

$$r = 3$$

$${}_7C_4 \cdot {}_6C_3 = [7!/(7-4)!4!] \cdot [6!/(6-3)!3!]$$

$$= (7!/3!4!)(6!/3!3!)$$

$$= (35)(20)$$

$$= 700 \text{ ways}$$

3. Combination in a Series

$${}_nC_1 + {}nC_2 + \dots + {}nC_r = 2^n - 1$$

Example 1.

In how many ways can a president of the class assign at most five of his classmates to clean the room every Friday?

Solution:

$$\begin{aligned} 2^n - 1 &= 2^5 - 1 \\ &= 32 - 1 \\ &= 31 \text{ ways.} \end{aligned}$$

Exercise 1.2 Permutations & Combinations

Solve the following problems completely.

A. Permutations

1. How many motor vehicle number plates can be made (a) for old plate numbers, if each plate contains 3 different letters followed by 3 different digits? ; (b) for old plate numbers, if the first digit cannot be zero? ; (c) for new standardized plate numbers, if each plate contains 3 different letters followed by 4 different digits? ; (d) for new standardized plate numbers, if the first digit cannot be zero? ; and (e) for motor bicycle plate numbers, if each plate contains 2 different letters followed by 5 different digits?

2. (a) Obtain the number of ways in which 4 students can sit in a row. (b) How many ways are there if two of the students insists on sitting next to one another?

3. How many ways can 5 candles of different colors be placed in a circular holder?

4. There are 7 letters in the word COMPANY. How many 4 letter words can be permuted of (a) different letters?, (b) consonants only?, (c) begin and end in a consonant?, (d) begin with a vowel?, (e) contain the letter C?, (f) begin with M and end in a vowel?, (g) begin with N and also contain Y?, and (h) contain both vowels?

5. How many different marks, each consisting of 7 asterisks marked in a letter can be made from 3 black, 1 red, and 3 blue asterisks?

6. Determine the number of permutations that can be formed from all the letters of each country's name: Mississippi, Tennessee, Alaska, and Philippines.

7. How many ways can 9 teachers be seated on a bench if there are only 5 seats available?

8. Determine the number of ways in which 7 stars of different sizes be hung in a circle with strings.

9. A basket contains 10 rose flowers. What is the number of ordered samples of size (a) 3 with replacement; (b) 3 without replacement; (c) 4 with replacement; and (d) 5 without replacement?

10. How many positive integers with 2 different digits in which 0 cannot be the first digit that are (a) less than 50?, (b) greater than 80?, (c) even?, (d) odd?, and (e) divisible by 2?

B. Combinations

1. Determine the number of ways can a group of 4 boys and 3 girls be selected from 6 boys and 5 girls?

2. The representative of 5 student leaders is chosen every semester to attend the leadership training. How many ways (a) if the representative can be chosen from 10 qualified students, (b) if two of the qualified students will not attend the training together, and (c) if two of the qualified students are twin and will only attend the training together?

3. A computer technician is to repair 5 out of 7 computers in the Laboratory room. (a) How many choices has he? (b) How many if he must repair the first 2 computers?

4. Find the number of ways a coach can make a choice of one or more players from 8 eligible players?

5. A group has 8 businessmen and 3 housewives. (a) In how many ways can the group manager choose a representative of 5? (b) How many of them will contain at least one housewife? (c) How many of them will contain exactly one housewife?

6. (a) In how many ways a sales lady with 15 associates can invite 7 of them to attend the party?, (b) if 3 of them are neighbors from far place and will not attend separately?, and (c) if 3 of them will not attend together?

7. The points N, O, P, Q, R, S, T, U, V, and W are in a plane for which only two are on the same line. (a) How many lines are determined by the given points? (b) How many of these lines do not pass through S or T? (c) How many quadrilaterals are formed by the given points? (d) How many of these quadrilaterals contain the point R? (e) How many of these quadrilaterals contain the side VW?

8. A person is to guess the content of 8 out of 11 small boxes. (a) How many guesses has he? (b) How many if he must guess the content of the first 2 boxes? (c) How many if he must guess the content of the first or second box but not both? (d) How many if he must guess exactly 3 the content of the first 5 boxes? (e) How many if he must guess at least 3 the content of the first 5 boxes?

9. A bingo player has 5 different cards. How many ways can he select of different cards?

10. There are 26 letters in the English alphabet of which 21 are consonants. How many 7 letter words can be formed (a) if there are 4 different consonants and 3 different vowels?, (b) if these contain the letter b?, and (c) if these contain the letters d and n?